

INFECTIOUS DISEASE & TRANSMISSION BASED PRECAUTIONS

There are 4 precautions:

1. Standard/Universal

- 2. Contact** → Private room preferred, cohort of same disease that are POSITIVE through CULTURE, no mask, gloves, gowns, hand washing, eye face shield no unless needed, special filter mask no, patient mask no, dedicated equipment (stethoscope, BP, etc, toy would be dedicated), negative airflow no.
- Anything ENTERIC which means can be caught from intestine (fecal-oral)
 - CDIFF
 - Hepatitis A (A stands for ANUS = fecal oral)
 - Cholera
 - Staph infections
 - RSV – respiratory syncytial virus which is fatal in babies
 - **How is it transmitted? DROPLET. But it is on CONTACT precaution. Because with little kids, they get it through contact of contaminated objects.**
 - Herpes
 - Shingles (herpes zoster)

- 3. Droplet** → Private room preferred, cohort of same disease based on POSITIVE CULTURE, mask, gloves, NO gown, hand washing, eye face shield not unless needed, special filter mask no, patient mask needed when leaving room yes, dedicated equipment yes, negative airflow no.
- Bugs that travel 3 feet
 - Meningitis
 - H flu which causes epiglottitis

- 4. Airborne** → Private room required unless cohorting, mask yes, gloves, gown not necessarily, hand washing, eye face shield not unless needed, special filter mask only for TB, patient mask needed when leaving room yes, dedicated equipment not necessarily, negative airflow yes.
- Measles
 - Mumps
 - Rubella
 - TB
 - **Spread by droplet, but airborne precaution.**
 - Varicella chicken pox

PPE DONNING AND DOFFING

- Always take it **off** in alphabetical order
 1. Gloves
 2. Goggles
 3. Gown
 4. Mask
- Always put **on** in reverse alphabetical for G's and MASK is second
 1. Gown
 2. Mask
 3. Goggles
 4. Gloves

MATH PROBLEMS

A. Dosage calculations

B. IV drip rates:

Formula: $\frac{\text{Volume} \times \text{drop factor}}{\text{Time in minutes}}$

Minidrip/microdrip: 60gtts/ml

Macro drip: 10gtts/ml

C. Pediatric Dose Questions

2.2 lbs per 1 kg

Amount per day vs amount to be given at one time

D. IV Replacement Questions

Do you use leading 0's? YES, as long as it maintains place. 0000.0003000

Boards will tell you what to round to. Tenths/hundreths.

You do not have to write units. Just the numbers.

CRUTCHES CANES AND WALKERS

CRUTCHES

A) How do you measure the length of crutch?

- 2-3 finger widths below the anterior axillary fold to a point lateral to and slightly in front of the foot.
- You don't measure any landmark on the foot and NOT on the axillae. Rule these out!

B) How to measure the hand grip?

- When the handgrips are properly placed, the angle of elbow flexion is about 30 degrees

C) How to teach crutch gaits. There are four.

- **Two Point**
 - You move a crutch and the opposite foot together followed by the other foot and crutch
 - Two together at every time
- **Three Point**
 - You move two crutches and bad leg together
 - Three things move together at every time
 - 3 together then 1
- **Four Point**
 - You move everything separately
 - You have two legs and two crutches
 - You move ANY crutch, but once you move that, your sequence is LOCKED IN
 - You then move the opposite foot followed by the other crutch followed by the other foot
 - Very slow but very stable
- **Swing Through**
 - Non weight bearing (amputations)
 - You plant the crutches then you swing through
 - You never put the leg down

D) When do they use these?

- "Even for even, odd for odd" Therefore, ask yourself HOW MANY LEGS ARE AFFECTED?

1. Use the even numbered gaits (two and four) when the weakness is evenly distributed (when you have even number of legs messed up).

- Use two point for mild problem (mild bilateral weaknesses).
- Use four point for severe problem (severe bilateral weaknesses).

2. Use the odd numbered gait (three point) when one leg is odd.

3. If they can't bear weight / amputation for instance, use swing through.

Examples:

1. Early stages of rheumatoid arthritis. Two point → systemic disease therefore two.
2. LAK amputation. Swing through.
3. First day post op right knee replacement, partial weight bearing allowed. Three point.
4. Advanced stages of amyotrophic lateral sclerosis. Four point.
5. Left hip replacement, second day post op, NWB. Swing through.
6. Bilateral total knee replacement, first day post op, WB allowed. Four point.
7. Bilateral total knee replacement, three weeks post op. Two point.

E) Going up and down stairsCRUTCHES

- "Up with the good, down with the bad"
- Upstairs: lead with good foot then crutches go second.
- Downstairs: lead with the bad foot then crutches go second.
- Crutches always move with bad leg.

F) MiscellaneousCANES

- Always hold on opposite side of bad leg/GOOD leg side, but advance it with the bad leg

WALKER

- Pick them up, set it down, walk to it
- If they need to tie their belongings to it, tie to side not at the front → tip the walker over
- Boards does not like wheels on walkers or tennis balls on walkers

DELUSIONS, HALLUCINATIONS, ILLUSIONS

NON PSYCHOTIC VERSUS PSYCHOSIS

- You must decide whether the patient is non psychotic or psychotic when you receive a psychiatric question!
 - This matters because determines treatment, goals, meds, prognosis, LOS, legalities, etc.

1. A non-psychotic person has insight and is reality based.

- Yes they are emotionally ill but they are NOT psychotic.
- Insight: They know what's wrong with them. They know they have a problem. They know how it's messing up their life.
- Reality Based: Their senses (eyes, ears, mouth, nose, touch) MAKE sense.
- **Treatment:**
 - **Good therapeutic communication**
 - Talk about how they feel
 - It is the RIGHT answer that would be RIGHT for every other type of patient who is not psychotic (med surg/community/OB/etc.)
 - Tell me more...
 - That must be very difficult...
 - How are you feeling right now...
 - What do you mean by...
 - Etc.
- **Example**: I am depressed and I don't feel like doing anything.
 - She is non psychotic because she knows she's depressed.
 - How are you feeling right now?
 - What is currently causing distress for you?
 - How much energy do you have today?
 - Etc.
- Therefore, these type of questions are NOTHING SPECIAL.

2. A psychotic person has no insight and is not reality based.

- They think everyone else is sick and everyone else has a problem. They blame everybody else.
- Insight: They don't know they are sick. They may be able to state the disease but they don't know how it is affecting their life.
- Not Reality Based
- **Treatment:**
 - Good therapeutic skills do not work with these patients.
 - Unique specific strategies are needed.

SYMPTOMS OF ILLUSIONS/HALLUCINATIONS/DELUSIONS

- Psychotic symptoms
- ONLY psychotic persons HAVE these symptoms
- **Example:** Alice is depressed she said, "I can't stand this depression, it's ruining my life."
 - Dec energy level – yes
 - Psychomotor retardation – yes
 - Delusions of persecution – no
- **Delusion:** False fixed (they don't change it) idea or belief. There is no sensory component. With a delusion you're not hearing/tasting/seeing anything. You're ONLY thinking it. It's just a thought.
 1. **Paranoid delusion:** False fixed belief that people are out to harm you. The police/mafia/wife/kids lying/stealing/etc.
 2. **Grandiose delusion:** You think you are superior. Christ/ghandi/smarest/etc
 3. **Somatic delusion:** False fixed belief about a body part. I have x-ray vision, I can melt stones with my eyes, my brain is a martian super conducting proton accelerator. There are worms inside my arm. Pregnant 83y/o male.
- **Hallucination:** False fixed sensory idea or belief. You hear/feel/taste/smell/touch these things.
 - Five types of hallucinations; one for each sense.
 - Most common is **AUDITORY**.
 - Most common auditory hallucinations are voices telling you to hurt yourself.
 - Next most common is **VISUAL**.
 - Third most common is **TACTILE**.
 - Last two are **GUSTATORY (taste)** and **OLFACTORY** are relatively rare.
- **Illusion:** Misinterpretation of reality. It is a sensory experience.

DIFFERENTIATION BETWEEN A HALLUCINATION AND ILLUSION

- With an illusion there is a referent in reality.
 - Referent: Something to which a person refers to when they say something
 - There's actually something there, they just misinterpret what it is
- With a hallucination, there's absolutely nothing there.
- **Examples:**
 - "Listen, I hear demon voices." This is an example of an auditory hallucination because there was nothing there.
 - A client overhears nurses and doctors laughing and talking at the nurses station. "Listen, I hear demon voices." This is an example of an illusion because they were talking, but it was not demons.
 - A client stares at the wall and says, "Look, I see a bomb." Hallucination.
 - A client stares at the fire extinguisher, "Look, I see a bomb." Illusion. Because there is a referent, yes it was the fire extinguisher.

HOW DO YOU DEAL WITH PSYCHOTIC SYMPTOMS?

- Ask yourself, what is there problem? What kind of psychosis do they have?
 - There are basically three types of psychosis
1. **FUNCTIONAL PSYCHOSIS:** They can function in every day life. They can have a marriage, family, job, etc. NO brain damage, just chemical imbalance & ineffective coping.
 - a. **Four** diseases that fall under 90% this “**schizo, schizo, major, manic**”
 - i. **Schizophrenia**
 - ii. **Schizo-affective disorder**
 - iii. **Major depression (different from depression)**
 - iv. **Manic**
 - Are bipolars functional? Yes. Are they always psychotic? No. Only in acute phases.

2. **PSYCHOSIS OF DEMENTIA:** Actual brain damage occurs → psychosis
 - a. Alzheimers
 - b. Stroke
 - c. Organic brain syndrome
 - d. “senile”
 - e. “dementia”

3. PSYCHOTIC DELIRIUM

DEALING WITH PSYCHOTICS

FUNCTIONAL PSYCHOTICS

- No brain damage
- Therefore, potential to learn reality
- Nurse: Teach reality by the use of the four step process!

1. Acknowledge feeling.

- Usually the word “FEEL” is in the answer
- Or you can specify a feeling for instance
- I see you’re _____
- This must be very distressing for you

2. Present reality.

- I know that _____ is real to you but I do not _____
- Tell them what is reality, “I am a nurse, this is a hospital, etc.”

3. Set a limit.

- This topic is off limit
- We are not talking about this
- Stop talking about those _____
- We’re not going to talk about those voices
- You can be this directive/strong!

4. Enforce the limit.

- I see you're too ill to stay reality based so our conversation is over
- Ending the conversation
- Not taking away a privilege → punishment
- Bad answers construed as punishment → "Since you can't follow the rules, you lose your telephone privileges, etc"
- The only enforcement is ending the conversation!

Example: Schizophrenia says I'm going to kill you all tomorrow and slit your throats.

1. I see that you're upset.
2. We are all going to be kept safe while we're here. (POSITIVELY STATED) rather than, you are not going to kill anybody as this is inappropriate (NEGATIVELY STATED).
3. We are not going to talk about that kind of stuff. Those kinds of ideas are real to you but are off limits in this conversation.
4. I see you're too ill to have a reality based conversation, so our conversation is over. Would you like some medication to help you with this symptom?
 - a. Show them that this is not them it is their illness
 - b. They can take medication to deal with symptoms and be compliant with medications!

PSYCHOSIS OF DEMENTIA

- Structural brain damage
- Cannot learn reality
- Two step process

1. Acknowledge their feelings.

2. Redirect them.

- Channel them to do something they can do instead of what they can't do.

3. DO NOT PRESENT REALITY. ***

- Don't confuse this with **reality orientation** → which is ax their orientation and telling them person, place, time. **This is ALWAYS appropriate.**

4. DO NOT SET LIMITS.

Example: You have a patient with dementia who is psychotic. She is in the waiting room; in the lobby of the nursing room; all dressed up. It's Sunday. You say, "Ms. Smith you're all dressed up." She says, "Yes, my husband is gonna pick me up, we're going to church." Problem: Husband is dead for 10 years.

She has a: False fixed belief → **DELUSION**

What do you say to her?

- That sounds like an exciting thing to do.
- That sounds interesting.
- You don't have to ASK how she feels, you can just say...
 - Acknowledge: See how they feel and express it
 - Explore: Asking them

Redirect her:

- Why don't we sit down here and talk about what's going to happen at church today while we wait?
 - Reinforce memory, which is good with dementia → what do you like about church? Who's your pastor? Etc. then you can ask about family → etc.
 - Eventually, the patient will forget about the delusion.

PSYCHOTIC DELIRIUM

- Temporary, sudden, dramatic, secondary, loss of reality.
- Usually due to some chemical imbalance in the body.
- Different from functional:
 - It's temporary and sudden
- Different from dementia:
 - It's temporary, sudden, secondary
- Who are these people?
 - People who are crazy for the short term because of something else that's causing them to be crazy
 - For instance, drug reaction → lose touch with reality
 - People that are high on uppers
 - Withdrawing from downer (DT would be this)
 - Post op psychosis especially in OA
 - ICU psychosis due to sensory deprivation
 - UTI in OA
 - Thyroid storm
 - Adrenal crisis
- Good news: It's temporary
- Treatment:
 - Removing underlying cause
 - Keep them safe
- Two step process

1. Acknowledge feeling.

2. Reassure.

- That it's temporary
- They will be kept safe

3. DO NOT PRESENT REALITY.

4. DO NOT REDIRECT THEM BECAUSE IT WON'T WORK.

EXAMPLES THAT DIFFERENTIATE BETWEEN 3 TYPES OF PSYCHOSIS

1. Person with schizo affective disorder who points to two people across the room and they say, "Those people are plotting to kill me". What would you say?

Schizo affective → FUNCTIONAL psychotic.

Therefore, follow FOUR step process.

1. I see that you are frightened.
2. Those people are not planning to kill you, we are all safe.
3. Furthermore, we are not going to discuss this.
4. If they keep talking about this, I see you're too ill to have a conversation right now.

2. Person with Alzheimer's who points to two people across the room and they say, "Those people are plotting to kill me". What would you say?

Dementia → DEMENTIA psychotic.

Therefore, TWO step process.

1. I see that you are frightened.
2. Why don't we go somewhere where you can feel safe.

3. Person with delirium tremens who points to two people across the room and they say, "Those people are plotting to kill me". What would you say?

Delirium tremens → DELIRIUM psychotic.

Therefore, TWO step process.

1. I see that you are frightened.
2. This is temporary, you are safe here. This feeling will go away when you get better.

MISCELLANEOUS

- **Real Sick Personality D/O:** Antisocials, Borderlines, Narcisists can treat like FUNCTIONAL psychotics.
 - Most of the others → good therapeutic communication.

LOOSENING OF ASSOCIATION: Your thoughts are all over the map.

1. **Flight of Ideas** – You go from thought to thought to thought. You say phrases that are coherent but the phrases are not tightly connected. Each phrase by itself is coherent but together they are not.
2. **Word Salad** – They cannot make a phrase that is coherent. They just babble random words. They are sicker than flight of ideas.
3. **Neologism** – Making up imaginary words. You're a blinsabik. You're a slaboshnizak.
4. **Narrowed Self Concept** – When a **psychotic** refuses to leave their room or change their clothes. It is a functional psychotic. The reason why they are doing it is because the way they define who they are is very narrow → based on two things → where they are what they are wearing.
 - a. Do not make a functional psychotic get dressed → panic
 - b. Use four step process
 - c. If not a psychotic → Use therapeutic communication skills and be directive! Just like if you were dealing with a post-op
 - d. The only time you're allowed to make decisions for patients are for those who are depressed but are not psychotic
5. **Ideas of Reference** – When they think people are always talking about them.

DIABETES

Diabetes: Cannot metabolize glucose → cell death because glucose is needed for energy

Diabetes Insipidus:

- is a total different disease; not a type of diabetes mellitus
- Diabetes Insipidus is polyuria, polydipsia → dehydration due to low ADH
- This looks like a lot like diabetes mellitus but they're not the same thing
- It is like diabetes mellitus with the fluid part but not the glucose part
- Do they have a low urine output or a high urine output? HIGH just like diabetes mellitus
- Opposite of diabetes insipidus → SIADH
- Everybody knows DM has polyuria and polydipsia → DI also has the same symptoms → SIADH is the opposite = oliguria and are not thirsty because they are retaining water

Relationship between urine output and urine specific gravity:

- High urine output = Low urine specific gravity
- Low urine output = High urine specific gravity
- Inverse relationship

Questions/Examples:

- DM: high urine output, low urine specific gravity
- DI: high urine output, low urine specific gravity
- SIADH: low urine, high urine specific gravity
- **Of the three, who would have FVD?** DM, DI.
- **Of the three, who would have FVE?** SIADH.

Type I versus Type II Diabetes Mellitus:

Type 1:

- Three names:
 - Insulin Dependent,
 - Juvenile Onset,
 - Ketosis Prone
- Juvenile onset = not used

Type 2:

- Opposite names
 - Non Insulin Dependent
 - Adult Onset
 - Non Ketosis Prone
- Adult Onset not used anymore

Signs and Symptoms

- Three P's
- **Polyuria:** Increased urine output
- **Polydipsia:** Increased thirst
- **Polyphagia:** Increased swallowing
 - Does not mean eating a lot
 - It can be a symptom after a thyroidectomy!!!

- It's only means eating a lot in diabetes. In every other context = swallowing.

Treatment

- If you don't treat DM1 they could **D.I.E.**
 - Diet → least important of the three
 - Insulin*** → most important tx modality
 - Exercise
- **DM2** they could **D.O.A.**
 - Diet → most important of the three
 - Oral hypoglycemic
 - Activity

Diet, Insulin, Exercise

- Diet is more of a focus for DM2

- a) It is a calorie restriction.
- b) They need 6 small feedings per day.
 - a. Keeps the blood sugar levelled → don't have big peaks → more normoglycemic

Question: You have a type 2 diabetic, what is the best dietary action to take?

- a. Restrict calories to an appropriate level.
- b. Divide their food into 6 feedings a day.

Thought process:

- "BEST" means you're only choosing ONE. Therefore, think it through.
- If you choose a, you will most likely eat three meals following their calorie restriction.
- If you choose b, you will split into 6 feedings, but they can eat as many calories as they want.
- Therefore, choose A.

Insulin

- Lowers the blood glucose
- Four types that you need to know

1. REGULAR: Humulin R, Novolin R, anything with R.

Onset: 1 hr

Peak: 2 hr

Duration: 4 hr

- It is clear in the bottle so it's a solution so it can be IV dripped
- Short rapid acting insulin (it was made before lispro that's why it's classified as short rapid acting)
- R sounds for Rapid and Run.

2. NPH:

Onset: 6 hr

Peak: 8-10 hr

Duration: 12 hr

- True intermediate acting insulin
- Cloudy → Suspension (not solution)
- Suspensions precipitates → particles fall to the bottom
- Therefore, NO IV drip → OD and brain death
- N stands for Not so fast (Intermediate) and Not in the bag.

1, 2, 4, 6, 8, 10, 12

124 = regular

681012 = NPH

They test the **PEAKS!** 2 and 8-10.

3: Humalog: Lispro

Onset: 15 mins

Peak: 30 mins

Duration: 3 hrs

15, 30, 3.

- Give as they begin to eat; give WITH meals
- Not ac meals

4: Lantus: Glargine

Onset:

Peak: none.

Duration: 12-24 hours

- So slowly absorbed it has no essential peak
- This one has low risk for hypoglycaemia (little to no risk)
- Only insulin you can safely give at bedtime

Check your expiration dates on insulin.

What action by the nurse invalidates the manufacturer's expiration date on the bottle?

- Opening it.
- The minute you open a vile, the expiration date is not valid.
- The new expiration date is 30 days after that!
- Make sure you write the date with EXP or OPENED then the date.

Refrigeration is optional. You don't have to refrigerate in the hospital but at home, the patient does.

- In the hospital → unopened viles should be refrigerated.
- When you open the vile → it now does not have to be refrigerated.

Exercise potentiates insulin. Does the same thing as insulin.

- Think of exercise as another shot of insulin
- Bring a rapid metabolizing carbohydrate snack

Sick Days

- Flu, diarrhea, otitis media, etc.
- When sick, their glucose goes up.
- Take insulin even though they're not eating.
- Stress causes this.
- Take sips of water because → dehydration
- Stay as active as possible to lower the glucose.

Complications of Diabetes

There are 3 acute complications, which you have to know!!!

1. Low blood glucose in DM1 or DM2 → insulin shock, hypoglycaemia, hypoglycaemic shock

Causes:

- Not enough food
- Too much insulin/medication*
 - o PRIMARY CAUSE → it is impossible for a diabetic to have a low glucose without being overmedicated.
- Too much exercise

Danger: Brain damage, which is permanent.

Signs and Symptoms:

- "DRUNK IN SHOCK"

DRUNK:

- Staggering gait
- Slurred speech
- Poor judgment
- Slow reaction time/delayed
- Labile emotions (all over the place – laugh cry, laugh cry)
- Decreased social inhibition (loud, obnoxious)

SHOCK (vasomotor)

- Hypotension
- Tachycardia
- Tachypnea
- Skin is cool and clammy
- Pallor
- Mottled extremities

Treatment:

- Rapidly metabolized carbohydrates (sugar)
 - o Any juice
 - o Regular pop
 - o Chewed up candy
 - o Milk

- Honey
- Icing
- Jam
- Jelly
- But you want a sugar + starch OR protein (ideal snack)
 - Crackers
 - Slice of turkey
 - Milk is sugar and protein (but you use SKIM milk so you don't break down the fat → ketones)
- If unconscious → GLUCAGON IM, Dextrose IV (D10 or D50)
 - Both are appropriate, what determines which route?
 - The setting

2. DKA = High BGL in DM1, diabetic coma

- Only type 1 because another name is ketosis prone

Causes:

- too much food
- not enough medication
- not enough exercise
- none of those are the #1 cause

#1 cause is: acute viral upper respiratory infections within the last two weeks.

- viral pharyngitis
- they recover within 3-5 days but after they recover initially, they start going downhill → lethargic
- Therefore, ASK IF THEY HAVE HAD AN INFECTION.
- The stress of the illness was not shut off → fats were burned for fuel

Signs and Symptoms:

- "DKA"
- Dehydration
 - Dry, poor elasticity, warm skin (water in body is same as water in car meaning it is a coolant), headache, flushed, tachycardia
- K for:
 - Ketones
 - Ketones in blood yes?
 - If you have ketones in your urine do you have DKA? Not necessarily.
 - Therefore ketones in blood confirms the diagnosis.
 - Kussmaul respirations
 - Deep and rapid
 - K⁺
 - High potassium
- A for:
 - Acidotic (metabolic acidosis)
 - Acetone breath

- Fruity odour
- Anorexia due to nausea
 - They don't want to eat

Treatment:

- DEHYDRATION → FAST IV fluids
 - IV with regular insulin at around 200ml/hr
 - Main solution doesn't matter even if it has D5W (because 1L only has 300 calories in it)
 - D5W does not even stay in the vein → it will not create hyperglycemia unlike D10 or D50

3. HHNK, HHS, HHNS → High BGL in DM2

- "NON KETONIC" → DM2
- Very simple to understand
- **This is the exact same thing as dehydration**
- Number one nursing dx: Fluid volume deficit
- Number one tx: IVF
- Outcomes: increased urine output, moist mucous membranes, etc.
- It's easy! It's DEHYDRATION.
- It's DKA without the K and the A. It's dehydration.

Questions:

Which one is insulin the most essential in treating? **DKA.**

Which one has a higher fatality rate? **HHNK/HHS.**

If a DKA or HHS comes in ER, which one is higher priority? **DKA.**

- DKA comes in a lot later due to worse symptoms that occur only later in disease
- HHNK is treated but in bad shape
- Die first without treatment: DKA

Long Term Complications of DM

Due to:

- Poor tissue perfusion
- Peripheral neuropathy

Renal failure, impotence, incontinence, can't feel when they injure themselves, can't heal properly when they injure themselves, retinal neuropathy.

Which lab test is the best indicator of glucose control? **HbA1C/glycosylated haemoglobin**

- In control: 6 and lower
- Out of control: 8 and above
- Therefore, at risk/on the border 7
 - Need evaluation, work up, infection somewhere.